

A Mathematical Specification for Multi-Image Projection, Alignment, Stereo Presentation, and Surface Reconstruction

Sightline Technical Reference

Abstract—This paper fixes the mathematical contract for a multi-image airborne imaging system that supports planar projection, sparse and dense correspondence, network attitude adjustment, uncertainty-aware reconstruction, surface products, reference-terrain registration, and stereo presentation. The specification separates physical geometry from plane-coordinate reparameterization and display-only transforms. It defines right-handed frames, full-source image coordinates, ray and plane equations, robust objectives, pass-correlated priors, tangent-space time-varying research, dense matching, multi-ray estimation, covariance, numerical precision, and degeneracy states. A direct procedural two-image red/cyan path is specified as the executable companion and first independent translation oracle. Detailed Jacobians, multi-image normal equations, dense costs, and conformance rules appear in the appendices.

Index Terms—photogrammetry, pushbroom imaging, bundle adjustment, stereo matching, anaglyph, triangulation, uncertainty, digital elevation model, numerical precision

I. SCOPE AND INVARIANTS

THE system considered here observes a physical scene from two or more time-dependent viewpoints. A view can be a frame image, a line-scanner product, or any source for which a continuous world ray can be formed at a full-source image coordinate. Measurements may be inspected pairwise, but accepted evidence belongs to one stable multi-image network. The mathematical contract therefore centers stable view identity, explicit frames and units, observable parameterizations, and immutable evidence generations.

Three kinds of transformation must remain distinct.

- 1) A *physical transform* changes a source origin, a source ray, a terrain surface, or another quantity that affects scientific geometry.
- 2) A *plane reparameterization* changes only the two basis vectors used to label points on an unchanged physical plane.
- 3) A *presentation transform* changes only camera or display position, eye parallax, visibility, or color composition.

Only the first category can change a reconstructed point or a backend scientific product. This distinction prevents a visually useful stereo exaggeration from being mistaken for a measurement correction.

All authoritative geometry, estimation, covariance, and procedural-reference operations use IEEE 754 double precision. Full-source coordinates are ordered as column then

row, $\mathbf{u} = [c, r]^T$, although arrays are indexed by row then column. Angles are radians and angular covariance is radians squared. Points and vectors are columns. These choices follow established photogrammetric and multiple-view practice [1]–[3].

II. FRAMES, TRANSFORMS, AND SENSOR FORMATION

A. Rigid transforms and active rotations

A right-handed rigid transform from frame b to frame a is

$$\mathbf{p}_a = R_{ab}\mathbf{p}_b + \mathbf{t}_{ab}, \quad T_{ab} = \begin{bmatrix} R_{ab} & \mathbf{t}_{ab} \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (1)$$

Composition follows the written frame chain, $T_{ac} = T_{ab}T_{bc}$. A small active rotation $\delta\theta$ left-composes a nominal attitude,

$$R^+ = \text{Exp}([\delta\theta]_{\times})R^0, \quad (2)$$

where, for $a = \|\delta\theta\|$,

$$\text{Exp}([\delta\theta]_{\times}) = I + \frac{\sin a}{a} [\delta\theta]_{\times} + \frac{1 - \cos a}{a^2} [\delta\theta]_{\times}^2. \quad (3)$$

The series limit is used near zero. Equation (2) is also the authoritative interpretation of an OPK increment: the named axes and order are declared once, never inferred from storage order.

B. Platform, gimbals, and line scanner

At acquisition time t and image column c , the source orientation is the ordered chain

$$R_{WI}(t, c) = R_{WB}(t)R_{BG_r}(t)R_{G_rG_p}(t)R_{G_pS}(t, c)R_{SI}. \quad (4)$$

The scanner term may vary by column. A body-frame lever arm ℓ_B gives the ray origin

$$\mathbf{g}_W(t, c) = \mathbf{p}_{WB}(t) + R_{WB}(t)\ell_B. \quad (5)$$

For calibrated image vector $\tilde{\mathbf{u}}$, the world ray is

$$\mathbf{v}_W(t, c, r) = \frac{R_{WI}(t, c)K^{-1}\tilde{\mathbf{u}}}{\|R_{WI}(t, c)K^{-1}\tilde{\mathbf{u}}\|}. \quad (6)$$

This model includes a linear pushbroom camera as a special case [4]. Timing, gimbal angle, lever arm, and calibration provenance are distinct inputs; one must not absorb their errors into an unlabeled image shift.

TABLE I
PRINCIPAL NOMENCLATURE. A SUBSCRIPT NAMES A FRAME; R_{ab} MAPS FRAME b COORDINATES INTO FRAME a .

Symbol	Meaning	Symbol	Meaning
W, E, B	World/project, local east-north-up, and platform body frames	G_r, G_p, S	Roll-gimbal, pitch-gimbal, and scanner frames
Π, C, D	Physical plane, presentation camera, and display frames	I_i	Full-source image frame for stable view i
$\mathbf{p}, \mathbf{g}, \mathbf{v}$	World point, ray origin, and ray direction	$\mathbf{q} = [x, y]^\top$	Coordinate in physical-plane basis E_Π
$\mathbf{u} = [c, r]^\top$	Continuous full-source column/row coordinate	$T_{ab}, R_{ab}, \mathbf{t}_{ab}$	Homogeneous transform, rotation, and translation
$\boldsymbol{\theta}$	Ordered omega, phi, kappa correction	$\delta\boldsymbol{\theta}$	Small local tangent rotation vector
$\mathbf{b}_{ij}$	Baseline from view i to view j	ρ, w	Robust loss and induced observation weight
J, N, Σ	Jacobian, weighted normal matrix, and covariance	$B_k(c), \mathbf{a}_{ik}$	Cubic spline basis and tangent control vector
d, λ	Disparity/search displacement and forward ray range	m	Explicit validity/evidence mask

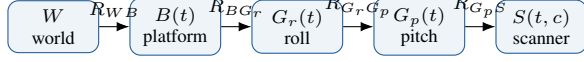


Fig. 1. The platform-gimbal-scanner order is explicit. Reordering factors changes the physical ray and is not a convention-only edit.

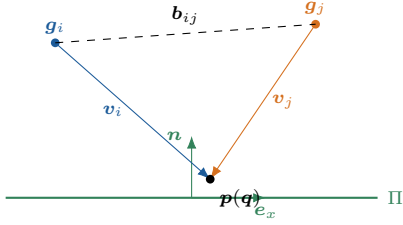


Fig. 2. A physical point is reconstructed from forward rays. The plane basis labels the point but does not alter either source ray.

III. PHYSICAL PLANES, RAYS, AND FULL-SOURCE PROJECTION

A. Plane construction and coordinates

A physical plane is

$$\Pi = \{\mathbf{p} : \mathbf{p} = \mathbf{p}_0 + E_\Pi \mathbf{q}\}, \quad E_\Pi = [\mathbf{e}_x, \mathbf{e}_y], \quad (7)$$

with $E_\Pi^\top E_\Pi = I_2$ and $\mathbf{e}_x \times \mathbf{e}_y = \mathbf{n}$. Coordinates and reconstruction are

$$\mathbf{q} = E_\Pi^\top (\mathbf{p} - \mathbf{p}_0), \quad \mathbf{p} = \mathbf{p}_0 + E_\Pi \mathbf{q}. \quad (8)$$

Rotating E_Π about $\mathbf{n}$ changes labels $\mathbf{q}$ but not Π . Tilting $\mathbf{n}$ or moving $\mathbf{p}_0$ changes the physical plane. Ordered corners use a declared clockwise or counterclockwise convention as viewed from $+\mathbf{n}$; the order is never reconstructed from array position.

B. Ray intersection

For ray $\mathbf{p}(\lambda) = \mathbf{g} + \lambda \mathbf{v}$, intersection with Π is

$$\lambda = \frac{\mathbf{n}^\top (\mathbf{p}_0 - \mathbf{g})}{\mathbf{n}^\top \mathbf{v}}, \quad \mathbf{p} = \mathbf{g} + \lambda \mathbf{v}. \quad (9)$$

The result is valid only when the denominator is separated from zero and $\lambda > 0$. Terrain intersection replaces the plane equation by a height or implicit-surface residual and uses a bracketed or otherwise guarded root solve. Voids, exclusions, horizon misses, and nonconvergence are named mask states.

C. Inverse mapping and resampling

The backend mapping is output driven. Source rays sampled on a full-source mesh produce pairs $(\mathbf{q}_{ik}, \mathbf{u}_{ik})$. A continuous inverse map is

$$\mathbf{u}_i(\mathbf{q}) = \mathcal{I}_{sc}(\{\mathbf{q}_{ik} \mapsto \mathbf{u}_{ik}\}, \mathbf{q}), \quad (10)$$

where interpolation is undefined outside supported topology. Radiometry is then sampled separately,

$$I_i^\Pi(\mathbf{q}) = \mathcal{I}_{img}(I_i, \mathbf{u}_i(\mathbf{q})), \quad m_i = m_{geom} m_{image} m_{operator}. \quad (11)$$

Nearest and bilinear sampling are explicit options. Invalid fill is an output policy, never evidence. Display pyramids or preview tiles cannot replace the full-source radiometric input.

IV. SPARSE CORRESPONDENCE AND TRACK EVIDENCE

Feature detection and description create candidate observations with scale, orientation, descriptor, mask, and source-coordinate provenance. Descriptor distance, ratio, uniqueness, tie, texture, and radiometric compatibility remain inspectable. Consensus fitting can use a similarity, affine, or other declared image-space model under a robust sampling method such as RANSAC [5]; descriptor construction may follow established local feature methods [6].

For physically calibrated rays, a baseline-normalized coplanarity residual is

$$e_{ij} = \frac{(\mathbf{v}_i \times \mathbf{v}_j)^\top \mathbf{b}_{ij}}{\max(\|\mathbf{b}_{ij}\|, \epsilon_b)}. \quad (12)$$

This scalar triple product is invariant to a uniform baseline scale. Degenerate baselines and invalid rays are reported, not assigned a small residual.

A track is a set of observations hypothesized to represent one physical point. It contains at most one observation from each stable view. Direct and composed paths are compared before merging; a cycle disagreement is diagnostic evidence, not another independent solver residual. The pair graph first spans every feasible connected component and then adds complementary loop chords only when they improve conditioning or validate an independent path.

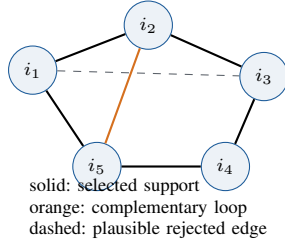


Fig. 3. Measurements are generated pairwise but reconciled into one network. Redundant pair multiplicity does not create independent physical evidence.

V. GLOBAL ATTITUDE ADJUSTMENT

A. Robust objective, priors, and gauge

Let δ collect the active attitude increments. With residuals $r_m(\delta)$, standard deviations σ_m , and prior factor L , the bounded objective is

$$\hat{\delta} = \arg \min_{\delta \in \mathcal{B}} \sum_m \rho\left(\frac{r_m(\delta)}{\sigma_m}\right) + \|L(\delta - \mu)\|^2. \quad (13)$$

Huber loss supplies an interpretable default robustifier [7]. Priors are outside observation robustification. Bounds are physical safeguards, and a bound hit is a diagnostic rather than proof of convergence.

Same-pass structure is

$$\delta\theta_i = \delta\theta_{p(i)} + \epsilon_i, \quad \sum_{i \in p} \Lambda_i \epsilon_i = \mathbf{0}, \quad (14)$$

where the second relation is a precision-weighted balanced gauge. Independent passes have separate common terms and are connected only by cross-pass evidence or an explicit prior. Silently fixing the first view is prohibited because it concentrates gauge uncertainty in an arbitrary observation.

At a linearization point,

$$N = J^T W J + L^T L, \quad \Sigma_\delta = N^\dagger, \quad (15)$$

only when rank, gauge, and conditioning justify covariance. The decomposition of data and prior objective, connected components, weak views, leave-one-edge sensitivity, and residual summaries by pass, time, and image region remain reported. This is the sparse block structure familiar from bundle adjustment [8].

B. Time-varying attitude research

After a constant solution, a bounded research model represents within-image attitude by local rotation vectors,

$$\delta\theta_i(c) = \delta\theta_{p(i)} + \sum_k B_{ik}(c) \mathbf{a}_{ik}, \quad (16)$$

where B_{ik} are open-uniform cubic B-splines initially spaced every 128 full-source columns [9]. Each attitude is composed through (2); Euler angles are never interpolated. A second-difference penalty $\sum_i \|D_2 \mathbf{a}_i\|^2$ supplies smoothness, and the differential controls retain a balanced pass gauge.

Post spacing doubles until local support, data rank, and condition are adequate, or until a declared maximum proves the model insufficient. A one-column model is only an

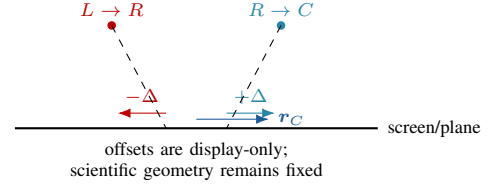


Fig. 4. Physical eye identity controls color; symmetric display translations control perceived separation and depth.

upper-bound experiment. Synthetic algebraic recovery does not prove physical observability, so time-varying corrections remain review-only until held-out physical evidence establishes stability and value beyond the constant model.

VI. PAIR CAMERA AND STEREO PRESENTATION

A. Pair viewpoint

A pair camera is a presentation object. Its target is the common-footprint centroid and its view direction points from a representative midpoint-derived camera position toward that target. The camera up vector is derived from the physical plane. Fitting, following, panning, zooming, and twisting do not change the plane, rays, matches, or reconstructed products.

B. Physical eye assignment with hysteresis

Let $\mathbf{r}_C$ be camera right and $\mathbf{g}_i$ representative source origins. The screen-horizontal positions are $s_i = \mathbf{r}_C^T \mathbf{g}_i$. Outside a normalized head-on threshold h , the smaller s_i is physical left and maps to red. If a previous assignment exists, it switches only when

$$\frac{s_{\text{previous right}} - s_{\text{previous left}}}{\|\mathbf{g}_2 - \mathbf{g}_1\|} < -h. \quad (17)$$

At an initially degenerate baseline, stable identity supplies the deterministic tie break. A manual swap is presentation state and never rewrites view identity.

C. Display-only parallax and red/cyan composition

For view width W_C , exaggeration η , base fraction β , and screen depth offset d_0 , define

$$\Delta = (\eta - 1)\beta W_C + d_0, \quad \mathbf{o}_L = -\Delta \mathbf{r}_C, \quad \mathbf{o}_R = +\Delta \mathbf{r}_C. \quad (18)$$

These offsets translate displayed eye surfaces only. On a fixed output plane, the equivalent inverse sampling query is

$$\mathbf{q}_i^D = \mathbf{q} - E_\Pi^T \mathbf{o}_i. \quad (19)$$

The physical $\mathbf{q}$, $\mathbf{p}$, plane, and source mesh remain unchanged. With unit grayscale Y_L, Y_R and common mask $m = m_L m_R$, the canonical companion anaglyph is

$$A(\mathbf{q}) = [Y_L(\mathbf{q}), Y_R(\mathbf{q}), Y_R(\mathbf{q})], \quad m = 1, \quad (20)$$

and explicit fill otherwise.

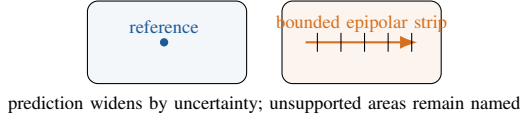


Fig. 5. Dense search is bounded by geometry and uncertainty, then audited by uniqueness, texture, and bidirectional consistency.

VII. DENSE CORRESPONDENCE

Dense processing is planned independently of the sparse alignment graph. Pair selection considers overlap, intersection angle, texture, radiometry, visibility, conditioning, expected work, and validation views. Sparse tracks may seed local disparity or direction priors, but unsupported regions remain explicitly unseeded rather than inheriting fabricated depth.

For zero-mean patches a, b , a common normalized cost is

$$C_{\text{ZNCC}} = 1 - \frac{a^T b}{\|a\| \|b\|}. \quad (21)$$

Gradient, census/rank, and phase-only costs are separate declared alternatives. Uniqueness, ties, low texture, prediction residual, and forward/backward consistency remain state evidence. If C_-, C_0, C_+ bracket a strict local minimum, bounded parabolic refinement gives

$$d^* = d_0 + \frac{C_- - C_+}{2(C_- - 2C_0 + C_+)}. \quad (22)$$

Left/right consistency rejects or labels an observation when $\|u_L - FB(u_L)\| > \tau_{FB}$. Semi-global matching is retained as a classical baseline, not an automatic universal choice [10], [11].

VIII. TRIANGULATION, UNCERTAINTY, AND SURFACE PRODUCTS

A. Pairwise and robust multi-ray reconstruction

Given unit rays (g_i, v_i) , the linear multi-ray point minimizes orthogonal ray distance,

$$\hat{p} = \arg \min_p \sum_i w_i \|(I - v_i v_i^T)(p - g_i)\|^2. \quad (23)$$

Weights are robust and equalized by unique view/pass evidence; pair multiplicity cannot increase independent support. Forward range, ray angle, normal rank, condition, residual, radiometry, visibility, and rejected-ray provenance determine reliability. Two-view triangulation is the labeled special case discussed by Hartley and Sturm [12]. Competing depth modes remain separate when the evidence supports more than one surface.

B. Covariance

With observation coordinates u , source-state variables s , and linearization Jacobians J_u, J_s ,

$$\Sigma_p = J_u \Sigma_u J_u^T + J_s \Sigma_s J_s^T + J_u \Sigma_{us} J_s^T + J_s \Sigma_{us}^T J_u^T. \quad (24)$$

Correlation terms are retained when errors share navigation, pass, or DEM sources. Numerical Jacobians use scale-aware perturbations and symmetry checks; analytic forms are preferred at frozen high-value kernels. An unavailable or rank-deficient covariance is labeled rather than replaced with zeros.

C. Fusion and surface formation

The authoritative point product preserves every contributing source observation and its full-source coordinate. Optional occupancy, Gaussian, mesh, and grid products are estimators over this evidence, not replacements for it. Sparse occupancy votes, uncertainty-aware splats, and direct multi-ray points can be compared over GSD-derived scale sweeps. A surface product records resolution, mask, uncertainty, contributing views/passes, competing modes, decimation, and whether it is authoritative or diagnostic.

IX. REFERENCE TERRAIN REGISTRATION

A terrain grid declares horizontal datum, vertical datum, units, accuracy, voids, masks, and transformations between geodetic, local ENU, and project world coordinates. A robust translation-only registration uses reference points d_k and normals n_k ,

$$\hat{t} = \arg \min_t \sum_k \rho \left(\frac{n_k^T (p_k + t - d_k)}{\sigma_k} \right). \quad (25)$$

The normal matrix includes point covariance, local terrain uncertainty, and a shared DEM error floor. Flat or weakly distributed terrain produces an ambiguity state. Buildings, water, voids, and caller exclusions remain named mask evidence. The method relates to point/surface registration principles [13], [14], but application is an explicit, atomic, and revertible physical source-origin change. Imagery-only points remain preserved for comparison.

X. NUMERICAL PRECISION AND CONFORMANCE

Large world coordinates can destroy small baselines if cast before local subtraction. The safe display boundary is

$$p_{\text{local}}^{(d)} = p_W^{(d)} - o_W^{(d)}, \quad p_{\text{display}}^{(s)} = \text{single}(p_{\text{local}}^{(d)}), \quad (26)$$

where superscripts identify double and single. Absolute world positions, ray intersection, inverse mapping, alignment, covariance, triangulation, and the procedural reference remain double. Dense costs or discardable display coordinates may use an explicit mixed boundary only after comparison with the double path.

Golden acceptance combines absolute, relative, angular, pixel, and mask terms. Every result records algorithm/version, precision, frame, units, input fingerprint, capability/fallback, deterministic options, and timing. A completed result must be invariant to cancellation polling and progress reporting. Independent ports compare plane/world coordinates, continuous source maps, values, masks, eye identity, presentation offsets, and stereo composition. Matrix rank and conditioning follow stable numerical linear algebra practice [15].

XI. LIMITATIONS AND DEGENERACIES

The model cannot recover information absent from the observations. Named stop or review states include parallel or behind-origin rays; empty or unverifiable source support; duplicate identities; feature ties; repeated or textureless regions; disconnected or ungauged networks; weak rank or condition;

bound hits; prior-dominated attitude; near-parallel triangulation; competing depth modes; terrain voids and datum ambiguity; and unavailable covariance. Presentation quality does not override these states.

Atmospheric refraction, rolling-shutter effects beyond the declared source model, cloud/water semantic screening, high-order calibration drift, and production time-varying attitude remain outside the authoritative scope until their parameters and observability are approved. GPU and lower-precision paths are optional accelerators. They cannot remove the CPU double reference or its error and mask semantics.

APPENDIX A

ROTATION AND INTERSECTION JACOBIANS

For $R^+ = \text{Exp}([\delta\theta]_\times)R$ near zero and fixed vector $\mathbf{x}$,

$$\left. \frac{\partial(R^+\mathbf{x})}{\partial\delta\theta} \right|_0 = -[R\mathbf{x}]_\times. \quad (27)$$

For the plane range in (9), define $\mathbf{a} = \mathbf{n}^\top(\mathbf{p}_0 - \mathbf{g})$ and $\mathbf{b} = \mathbf{n}^\top\mathbf{v}$. Then

$$\frac{\partial\lambda}{\partial\mathbf{g}} = -\frac{\mathbf{n}^\top}{\mathbf{b}}, \quad \frac{\partial\lambda}{\partial\mathbf{v}} = -\frac{\mathbf{a}}{\mathbf{b}^2}\mathbf{n}^\top, \quad (28)$$

and

$$\frac{\partial\mathbf{p}}{\partial\mathbf{g}} = \mathbf{I} + \mathbf{v}\frac{\partial\lambda}{\partial\mathbf{g}}, \quad \frac{\partial\mathbf{p}}{\partial\mathbf{v}} = \lambda\mathbf{I} + \mathbf{v}\frac{\partial\lambda}{\partial\mathbf{v}}. \quad (29)$$

The factors expose the $1/\mathbf{b}$ and $1/\mathbf{b}^2$ growth near parallel incidence. Such geometry must be rejected or downweighted before covariance is interpreted.

APPENDIX B

NETWORK LINEARIZATION AND OBSERVABILITY

At iteration k , robust weights are frozen while solving

$$\begin{bmatrix} W^{1/2}J \\ L \end{bmatrix} \Delta\delta = - \begin{bmatrix} W^{1/2}\mathbf{r} \\ L(\delta_k - \mu) \end{bmatrix}. \quad (30)$$

The data-only singular spectrum diagnoses observation geometry; the augmented spectrum diagnoses the numerical solve. A full augmented rank does not make an unobservable data mode observable: it may only show that a prior filled that mode. Reports therefore distinguish data rank, required gauge-adjusted rank, augmented rank, data condition, prior contribution, and prior dominance.

For time-varying controls, each observation row contains a pass-common block and at most four nonzero cubic basis blocks for its view. Coarsening reduces control count and increases support. Promotion requires held-out physical improvement, stable estimates under spacing perturbation, and an explicit source-geometry application contract.

APPENDIX C

DENSE COST AND CONSISTENCY DETAILS

For patches a, b with means $\bar{a}, \bar{b}$, ZNCC uses

$$\text{ZNCC}(a, b) = \frac{(a - \bar{a}\mathbf{1})^\top(b - \bar{b}\mathbf{1})}{\|a - \bar{a}\mathbf{1}\| \|b - \bar{b}\mathbf{1}\|}. \quad (31)$$

A gradient cost applies a robust norm to horizontal/vertical derivative differences. Census/rank costs compare local order

bit strings by Hamming distance. Phase-only cost normalizes cross-power magnitude before inverse transformation. Constant patches, zero denominators, saturated neighborhoods, and masks produce explicit states.

The subpixel denominator in (22) must indicate positive curvature and the offset must stay within a declared interval, normally one pixel. Forward/backward consistency is evaluated in continuous full-source coordinates. Chunked implementations include search and interpolation halos so interior results are independent of chunk boundaries.

APPENDIX D

MULTI-IMAGE ASSOCIATION, FUSION, AND REGISTRATION

Dense pair observations are reconciled by stable view identity before 3-D formation. A valid association has at most one observation per view; a conflict cannot be hidden by pair order. The normal matrix for (23) is

$$\mathbf{H} = \sum_i w_i(\mathbf{I} - \mathbf{v}_i\mathbf{v}_i^\top), \quad \mathbf{h} = \sum_i w_i(\mathbf{I} - \mathbf{v}_i\mathbf{v}_i^\top)\mathbf{g}_i, \quad (32)$$

with $\mathbf{H}\hat{\mathbf{p}} = \mathbf{h}$. Robust iteration updates residual weights while keeping unique view/pass accounting. The eigenstructure of $\mathbf{H}$ exposes weak ray geometry.

Surface fusion retains a reverse link from every output element to contributing points and source observations. DEM registration keeps vertical datum and frame transformations outside the optimizer. For translation-only registration, row k of $\mathbf{J}$ is $\mathbf{n}_k^\top$; a flat normal distribution makes horizontal translation weak regardless of point count. A shared DEM uncertainty component sets a covariance floor that cannot be averaged away as independent noise.

APPENDIX E

PROCEDURAL TWO-IMAGE COMPANION

The executable companion accepts two in-memory image arrays, one physical plane, camera values, two source-geometry functions, and explicit output and stereo options. Its visible order is:

- 1) create the output grid in $\mathbf{q}$ and reconstruct $\mathbf{p}$;
- 2) derive camera right and assign physical eyes;
- 3) form symmetric display-only offsets using (18);
- 4) intersect each source ray mesh with the physical plane;
- 5) construct the continuous inverse maps in (10);
- 6) query at (19), sample full-source radiometry, and mask;
- 7) compose (20); and
- 8) return double values and portable explanatory metadata.

It has no graphics, object hierarchy, hidden state, cache, tile pool, or implicit cast. Runtime source functions are not retained in output. Golden tests compare the direct algebra against production components rather than calling those components from inside the companion.

APPENDIX F

CROSS-LANGUAGE CONFORMANCE MATRIX

An implementation is conformant only when it passes the following layers:

- 1) frame, plane, ray, and inverse-map unit fixtures;
- 2) procedural two-image value, coordinate, mask, eye, and stereo fixtures;
- 3) sparse association and robust network fixtures, including degeneracy;
- 4) dense cost, subpixel, consistency, and chunk-boundary fixtures;
- 5) pair and multi-ray point/covariance fixtures;
- 6) surface and DEM registration fixtures; and
- 7) scale/precision fixtures through the required long-range geometry.

Each record names compiler/runtime, hardware, precision, acceleration, input fingerprint, tolerance, value error, mask mismatch, deterministic repeatability, and timing. Passing compilation or visual similarity alone is insufficient.

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